Volume of truncated pyramids and cones

1 $\frac{448}{3}$ cm^3

2 720 cm^3

3 263.89 cm^3

4

a

i 8π units^3 ii 27π units^3 iii 59.69 units^3

b

i 144π units^3 ii $\frac{1024\pi}{3}$ units^3 iii 619.94 units^3

Surface area of truncated pyramids and cones

5

a 6.32 cm **b** 231.7 cm^2

6

a

i 75π units^2 ii 27π units^2 iii 257.61 units^2

b

i 363π units^2 ii 147π units^2 iii 1212.65 units^2

c

i 135π units^2 ii 60π units^2 iii 603.19 units^2

d

i 585π units^2 ii 260π units^2 iii 2042.04 units^2

Other compound solids

7

a 169.65 cm^3 **b** 292.17 cm^3 **c** 221.74 cm^3 **d** 720.00 cm^3

8

a 173.13 cm^2 **b** 346.88 cm^2 **c** 245.04 cm^2 **d** 322.41 cm^2

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9 **a** Break the solid in to more familiar solids (e.g. a cylinder and two cones). Then, draw a net of each solid. Only include faces which are on the surface of the compound solid. Next is to calculate the area of each net. Finally, add up the area of all the nets.

b $193.07\ \text{cm}^2$

10

a $15.0\ \text{cm}$ **b** $916.72\ \text{cm}^3$

Applications

11

a $38\,475\ \text{cm}^3$ **b** $30\ \text{cm}$ **c** $11\,400\ \text{cm}^3$ **d** $27\,075\ \text{cm}^3$

12 $43.37\ \text{m}^3$

13

a $70.55\ \text{km}^3$ **b** $0.15\ \text{km}^3$ **c** $70.40\ \text{km}^3$

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